Agentic scientific curation

dynamic literature construction

predicate-state verification

DB-Curator turns open-ended scientific requests into auditable database entries, not just plausible answers.

Scientific need

Specialized databases are still built by experts who search papers, interpret context and record evidence row by row.

AI gap

LLMs can summarize papers, but database release needs candidate-level decisions and traceable evidence.

Core idea

Use agents as a controlled verification loop: retrieve, inspect, critique, and stop only when the rubric is satisfied.

What changes for scientists?

less manual literature triage

explicit inclusion/exclusion rules

database rows with provenance

human review focused on edge cases

Audience takeaway: the agentic part is not chat. It is dynamic corpus construction plus entry-level evidence checking.

How the agentic workflow works

The system builds the evidence set during the run, then verifies each candidate against a user-defined acceptance contract.

1. Scope

request → rubric
inclusion / exclusion

2. Literature

broad search
seed expansion
follow-up

3. Candidates

NER + grounding
ranking + queue

4. Verification

Researcher reads
Critique audits

5. Database

accepted rows
spans + IDs

Predicate-state loop

supported **unresolved** **exclusion triggered** **exhausted**

Critique asks: which required predicate is unsupported, which evidence is missing, and what should be searched next?

Application: release-ready database rows

Fixed 50-row cohorts test whether the system can produce useful entries, including valid compounds absent from existing benchmarks.

178

Known entries

Matched benchmark compounds after identity normalization.

68

Benchmark-missing valid

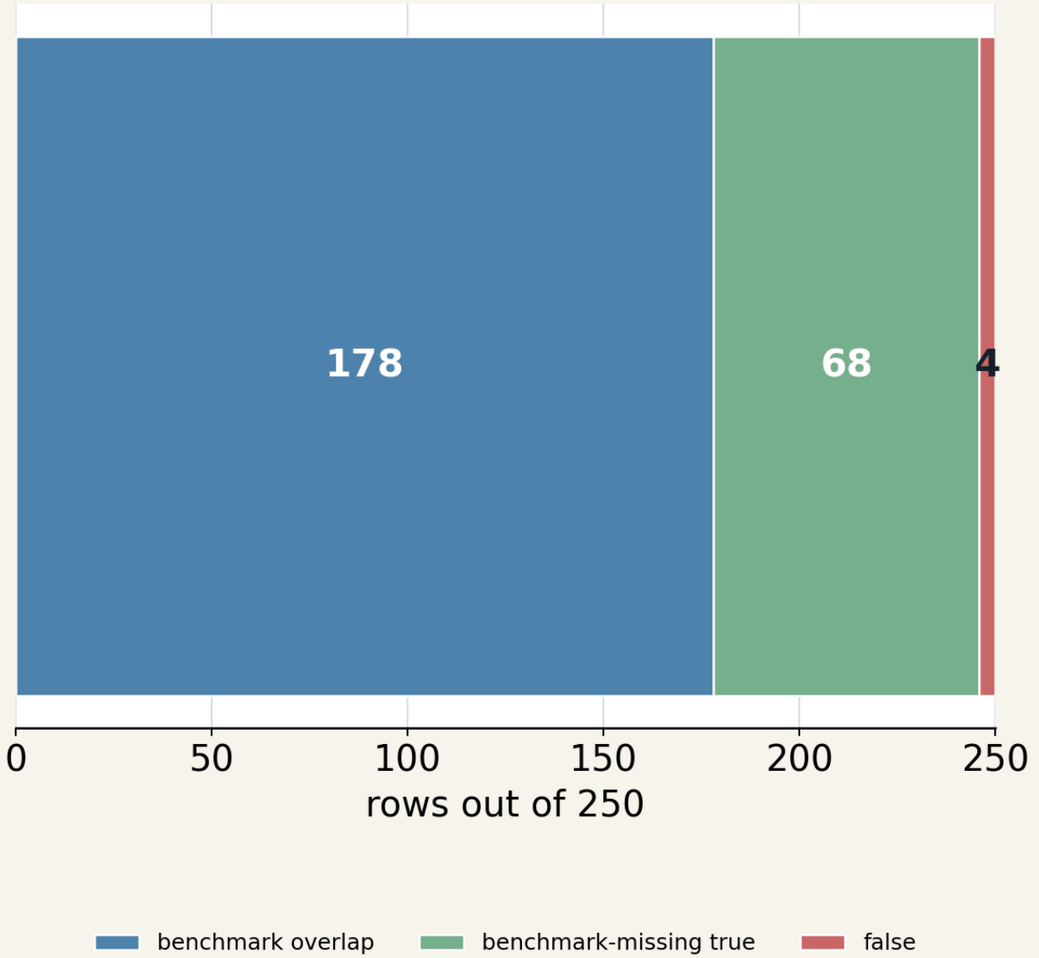
Non-overlap compounds judged valid by human experts.

4

Low release error

Non-overlap errors in the fixed-cohort audit.

Release audit over five domains



Scientific punchline: the system recovers known entries and surfaces literature-buried additions with auditable evidence.

Verification: evidence trails survive paper-level audit

The strongest result is not name generation. It is whether citations resolve to real papers and contain candidate-relevant support.

2,166/2,166

DB-Curator fixed-cohort audit

All cited primary-paper references contained the quoted span or equivalent local context.

11/257

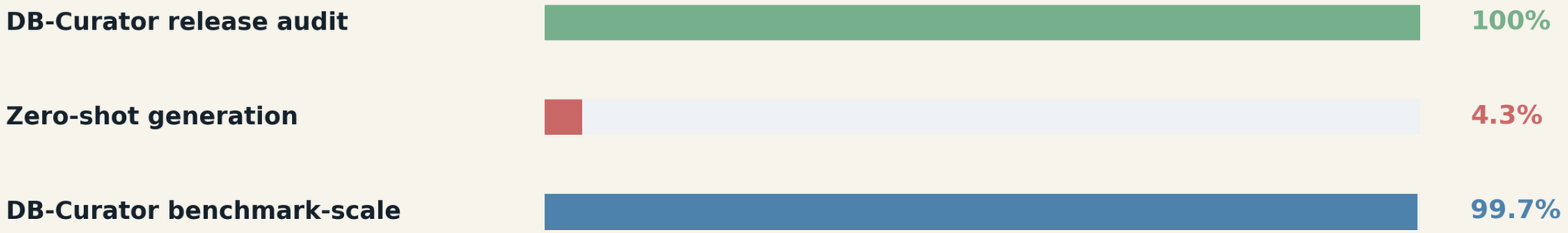
Zero-shot direct generation

The baseline produced plausible rows, but only a small fraction had usable content support.

7,430/7,453

Benchmark-scale audit

Across supplied candidates, DB-Curator maintained quote/context support for almost all cited references.



Verification rule: resolve paper identity → search recovered text → accept exact quote or equivalent local context → exclude database lookup pages.

Benchmark-scale verification across scientific domains

Five chemistry and biomedicine domains test whether agentic verification generalizes beyond one curated task.

1,887

Scale

Five scientific database domains.

977

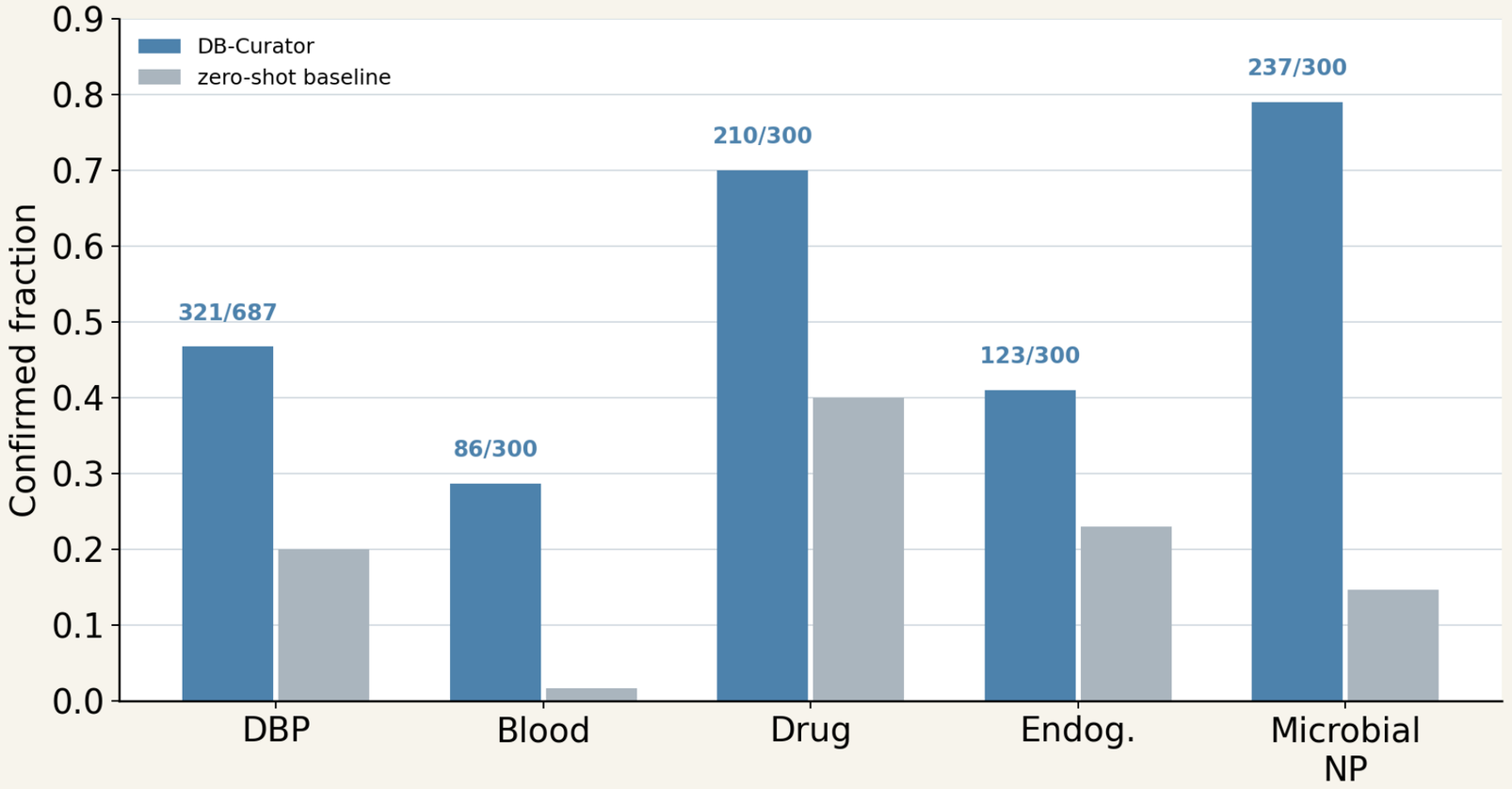
Confirmed

Evidence-supported entries.

375

Baseline

Direct labeling without retrieval.



ML takeaway: this is not one classifier threshold. Different topic definitions create different predicate gates and failure modes.

Why the agentic loop matters

Critique-guided retrieval changes decisions and leaves uncertain cases evidence-limited rather than hallucinated.

1. Researcher

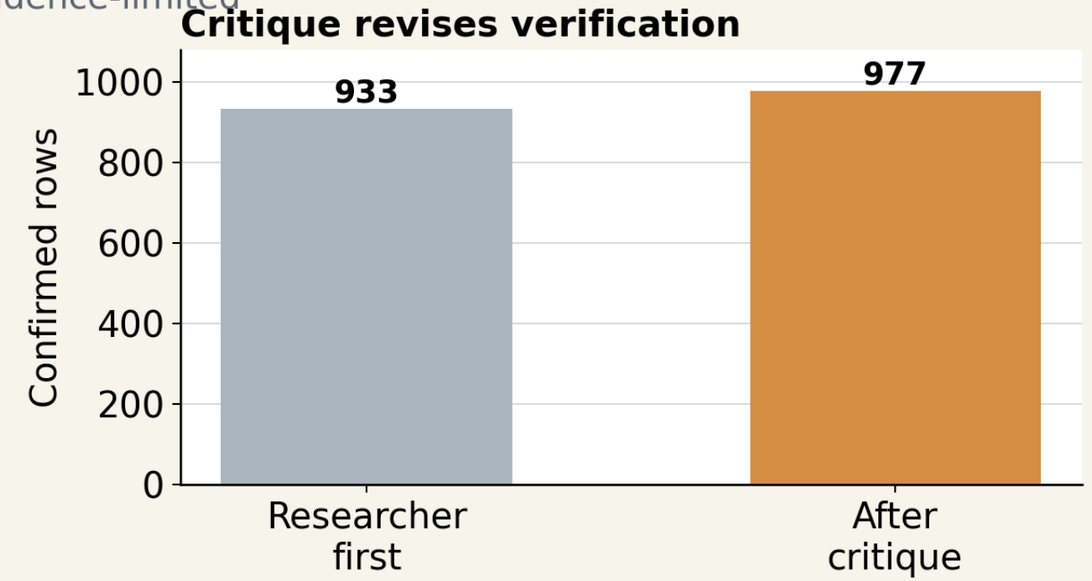
Finds papers, reads context and proposes a label with evidence.

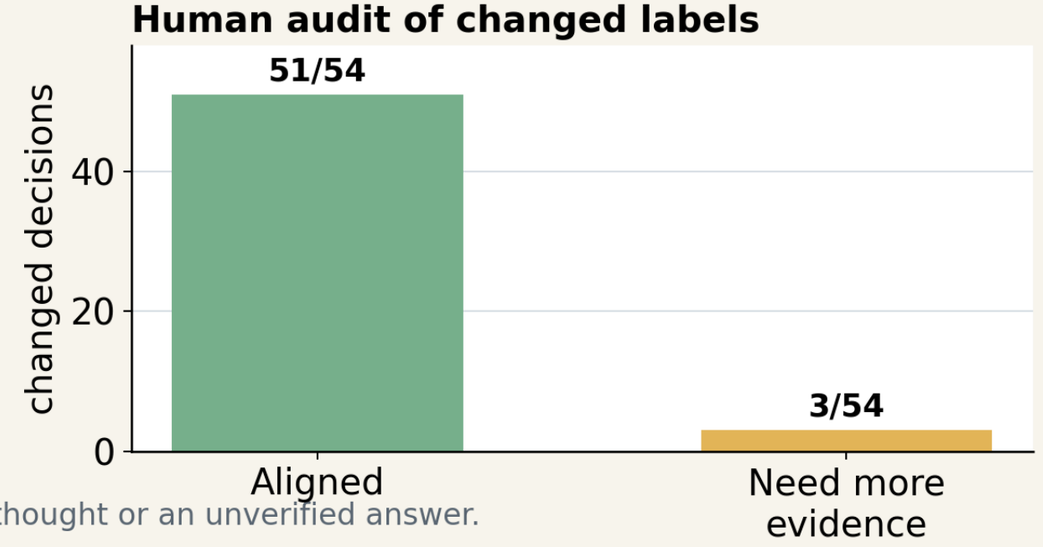
2. Critique

Checks the claim against rubric predicates and asks for targeted follow-up.

3. Terminal state

Accept only when supported; otherwise return evidence-limited or explicit negative.





Scientific takeaway: the user receives inspectable database rows, not a hidden chain-of-thought or an unverified answer.